



Basic mathematical model of overlimiting transfer enhanced by electroconvection in flow-through electrodialysis membrane cells

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ABSTRACT

Mechanisms for overlimiting current and concentration polarization in electrodialysis (ED) with ion-exchange membranes are not yet well understood despite its half-century history. A first-principles model involving the Nernst–Planck–Poisson equations fully coupled to the Navier–Stokes equations and containing no adjustable parameters is proposed. The calculated current–voltage (I – V) curve of an ED flow-through cell shows a linear region, a sloped plateau surpassing “limiting” current and a rapidly rising region characterized by increasing current oscillations. This curve and concentration profiles are compared with experimental data and with “classical” models. It is shown that the initial smooth region of the I – V plateau relates to a new electrokinetic mode, which is similar to the Dukhin–Mishchuk regime. The difference is in the fact that in the case of forced convection the tangential electric force producing stable electroconvection can appear at a homogeneous flat membrane due to nonuniformity in the lateral concentration distribution; membrane surface nonflatness or electric heterogeneity are not necessary. The oscillating region with a higher slope relates to the Rubinstein–Zaltzman unstable electrokinetic mode. I – V curves calculated with the no-slip and the Navier slip conditions are compared.

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1. Introduction

Electrodialysis (ED) is an electrically driven separation process using ion exchange membranes (IEM); it is largely applied during 60 years [1]. H. Strathmann characterizes ED as a mature technology, which still has technical and commercial limitations. There is a substantial need for basic as well as applied research to improve the processes involved [1]. In particular, intensive electric regimes are of great interest as they result in decreasing the area of expensive IEMs and lower investment costs. New ideas coming from other areas of knowledge, namely, from micro- and nano-fluidics [2], allows one to revise the conventional layout of ED and to propose novel conceptions of electro-membrane devices for water desalination using extended space charge as a barrier for ion transfer [3] or the effect of “deionization shock” in microstructures such as porous frits [4,5]. Relatively high voltages applied in these new devices give rise to phenomena similar to those occurring in intensive ED with IEMs, involving coupling of current and water transfer in solution.

Schematically, a conventional ED elementary cell is shown in Fig. 1. It includes an anion-exchange membrane (AEM), a cation-

exchange membrane (CEM) and solutions flowing between them. When an electric current passes through an ion-selective membrane separating two electrolyte solutions, salt concentration decreases at one membrane side and increases at the other due to selective transport of counterions in the membrane. In membrane science, the emergence of concentration gradients at membrane/solution interface induced by transmembrane driving forces is called “concentration polarization” (CP) [6]. Note that in electrochemistry, this term denotes the part of the polarization of an electrolytic cell resulting from changes in the electrolyte concentration due to the passage of current through the solution [7]. Here polarization is understood as the shift of the potential difference (PD) across the cell from its equilibrium value. When the term is used in this sense, it is equivalent to the other one, the “concentration overpotential” [8].

On the one hand, CP phenomenon results in solution desalination and concentration during ED. On the other hand, CP leads to higher energy consumption and to rate limitation of the process. Theoretically, under local electroneutrality (LEN) assumption, the current density, i , is limited by a value (the limiting current density, i_{lim}) that corresponds to zero electrolyte concentration at the surface of one of the membranes forming the desalting channel:

$$i_{lim} = \frac{DC_0F}{\delta(T_1 - t_1)} \quad (1)$$

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When studying 1D transport, δ is considered as a parameter, which can be evaluated from one or other reasons, for example, by fitting experimental $I-V$ curve. If we know i_{lim} , δ is readily found from Eq. (1). In 2D modeling under LEN assumption (the Nernst-Planck – Navier–Stokes (NP/NS) model) [11,12], the value of δ can be found as a function of the distance from the channel entrance, L ; the average Hagen–Poiseuille flow velocity through the channel, V_0 , and the inter-membranes distance, h , should be known. For a sufficiently small value of L ($< 0.02V_0h^2/D$), the approximation following from the L  v  que analytical solution [13,14] is valid [12]:

For such “short-length” channels, the bulk concentration remains equal to the inlet solution concentration; only the solution within the near-surface diffusion layers is the subject of variation.

The I - V curve calculated using the 2D NP/NS model [11,12] has a quasi-linear curve region at low concentration polarization, and a horizontal plateau corresponding to i_{lim} (Fig. 2). However, in practice, “overlimiting current” is always observed (Fig. 2). There are several mechanisms of the overlimiting conductance vividly discussed for decades [15–32]. They involve water splitting resulting in occurrence of new charge carriers, H^+ and OH^- ions [15,21]. In addition to the contribution of these ions in the transfer of electric current, they affect electric field in the depleted diffusion layer that causes an increase in the counterion current (the Kharkats effect of current exaltation) [20,21,23]. There are also two types of coupled convection [15,21], the gravitational one [21–27] and the electroconvection (EC) [17–19,21–32], and some other effects [32,33]. Experimental and theoretical evidences [17,22–31,34–36] show that in dilute solutions the overlimiting transfer is mainly due to electroconvection, which generally occurs as current-induced volume transport when an electric field is imposed through the charged solution [37].

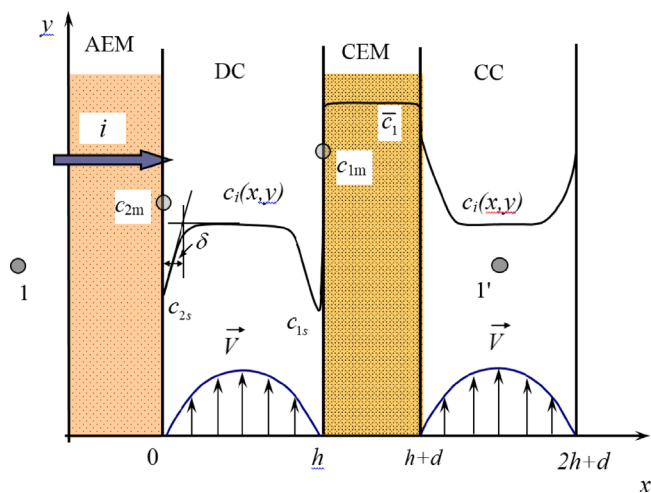


Fig. 1. Scheme of an elementary ED cell with an anion-exchange (AEM) and a cation-exchange (CEM) membranes; DC and CC are desalting and concentrating channels, respectively. Points 1 and 1' show the positions of equipotential planes parallel to the membranes and limiting the system under study.

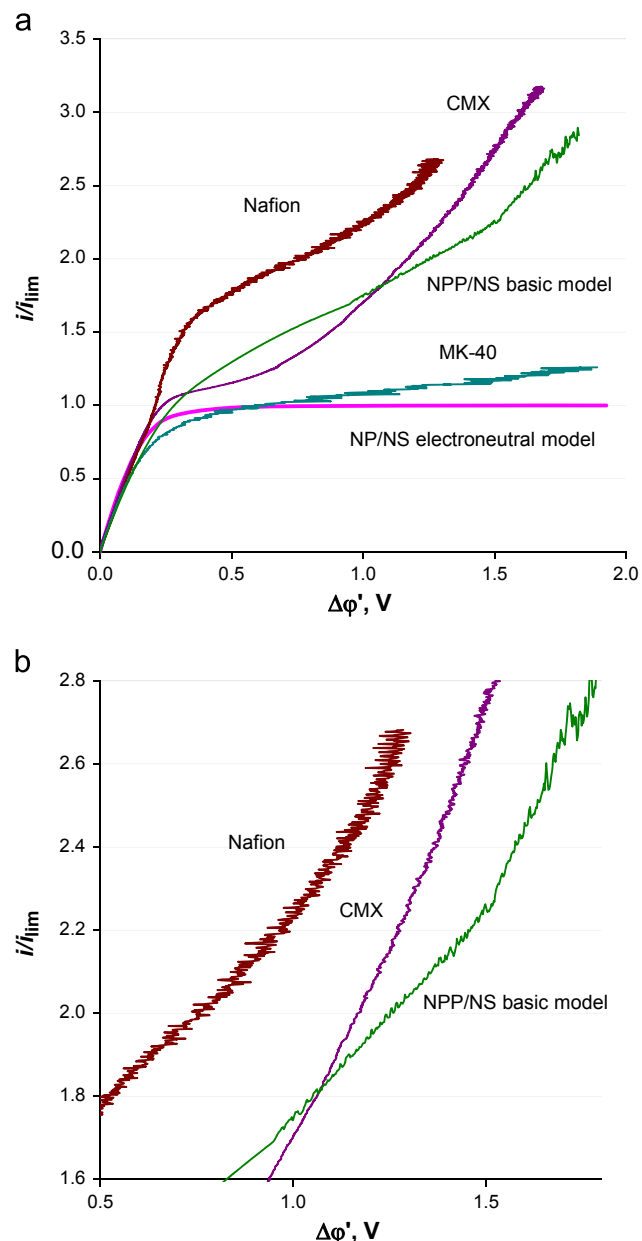


Fig. 2. Ratio i/i_{lim} as a function of the PD ($\Delta\varphi$) “corrected” for the ohmic losses in membranes [22]. Experimental data for different cation-exchange membranes (heterogeneous MK-40, homogeneous CMX and Nafion®-117) in 0.02 M NaCl solutions, and calculations according to the NP/NS model under the LEN assumption and to the “basic” NPP/NS model [Eqs. (3)–(26)]. Experimental curves were obtained with current sweep 0.01 mA s^{-1} . Calculation are made for a desalination flow-through ED cell, $h=1 \text{ mm}$, $L=2 \text{ mm}$, $V_0=0.8 \text{ mm s}^{-1}$, $c_0(\text{NaCl})=10^{-5} \text{ mol dm}^{-3}$, $\Delta\varphi_{\text{tot}}$ increased with 0.01 V s^{-1} rate; $N_a=N_c=1$. **a** is a magnification of **a**.

Physical and mathematical modeling of overlimiting transfer is a major challenge due to multi-physics and multi-scale nature of the problem. 1D models [23,38–40] inspired by the pioneer paper by Rubinstein and Shtilman [17] use the Poisson equation instead of “classical” LEN condition. It was shown that when $i \rightarrow i_{lim}$, the counterion concentration at the membrane surface does not tend to zero that occurs when applying the LEN condition. Instead, an extended space charge region (**SCR**) much larger than the equilibrium electric double layer (**EDL**) appears thus reducing the electroneutral DBL, which still controls the current transfer. However, the decrease in effective DBL thickness owing to the extension of the SCR is too small to explain the experimentally observed

growth in the overlimiting current density. To fit experimental I - V curves, δ entering into the boundary-value problem as a fitting parameter, must be adjusted as a function of i . Since in experiment, i continues to increase after reaching i_{lim} , one has to assume, according to Eq. (1), that δ progressively decreases with i [9,23]. However, the physics, which is behind these 1D models, cannot explain the effect of variation of δ or to give another plausible explanation of overlimiting ion transfer.

In fact, 2D modeling involving Navier–Stokes equations is needed to give a comprehensive description of overlimiting ion transfer. Dukhin and Mishchuk [41], and Rubinstein et al. [42] were the first who have given a qualitative explanation of overlimiting current by electroconvection. To produce electroconvective mixing of depleted solution resulting in essential increase of current above i_{lim} , two conditions should be fulfilled: (i) an extended SCR should be formed, far beyond the equilibrium EDL, and (ii) the distribution of electric field lines should be inhomogeneous to give rise to a spatially inhomogeneous electric bulk force. In the cases considered in [41,42], the second condition was realized due to spherical geometry of ion-exchange particles [41], or due to heterogeneous membrane surface [42]. Dukhin and Mishchuk has introduced the term “electroosmosis of the second kind” [41,43] to emphasize the difference between classical electroosmosis (of the first kind), which is the result of the action of the external electric field on the equilibrium EDL, and the studied here electroconvection caused by the action of the electric field on the extended SCR. Later on, also nearly at the same time, Dukhin and Mishchuk [44,45], and Rubinstein [46] have found analytical approximate solutions for the Navier–Stokes equations coupled to the Nernst–Planck–Poisson equation (NPP/NS) in ion-conducting heterogeneous systems: in solution/spherical ion-exchange particle [44] and solution/heterogeneous ion-exchange membranes [45,46].

In 2000, Rubinstein and Zaltzman [18] studied mass transfer in a solution situated between two homogeneous ion-exchange membranes without flowing through. They found a numerical solution for NP/NS equations under the LEN assumption in the entire system except for the boundary SCR. A special slip condition deduced by applying an asymptotic theory of non-equilibrium EDL was set on the boundary of the LEN region. They have shown that surface heterogeneity is not the necessary condition for rising electroconvection. A characteristic feature of this system is its hydrodynamic instability at sufficiently high voltages. A fluctuation in velocity or concentration leads to emergence of a spatially inhomogeneous electric field resulting in occurrence of vortices, which, in turn, contribute to electric field inhomogeneity.

A similar approach was applied by Dydek et al. [47] and by Green and Yossifon [48] who used the slip condition at micro-channel walls in the form of the Helmholtz–Smoluchowski formula. Note that in papers [17,47,48] as well as in other theoretical papers treating membrane or micro-nanofluidic systems at overlimiting currents [4,49–53], the cells under study involve no forced convection that does not conform to the practice of electrodialysis. In Refs [21,54] the forced convection was taken into account together with electroconvection. However, the coupling between the Navier–Stokes and the Nernst–Planck equations was realized only partially. The value of the electric body force used in the NS equation was approximately calculated by using Rubinstein's 1D model [17] for a given i and an appropriate value of δ . The numerical solution of the NS with body force allowed the authors to compute the structure of electroconvective vortices. The same approach was applied by Zabolotskii et al. [55] to heterogeneous membranes, for which the size of conducting and nonconducting surface segments was carefully evaluated.

Kwak et al. [30] recently studied, theoretically and experimentally, the balance of forced convection (produced by Hagen–Poiseuille

shear flow) and electroconvection flows in an ED channel between an anion- and a cation-exchange membranes. Among numerous findings of this study, there is deviation from the pair symmetry of EC vortices and the tendency to formation of a unidirectional vortex structure, as well as a new scaling law establishing the vortex size as a function of potential drop and the forced flow velocity. A good accordance between experimental and theoretical results is found, in particular concerning the direction and the size of EC vortices.

This short review shows that there are quite elaborated models involving coupling between the Nernst–Planck–Poisson and Navier–Stokes equations. However, most of the models concern the case where no forced convection occurs. The role of forced convection is not sufficiently well understood, in particular, in regard to the connection between the character of the vortex structure and the shape of the CVC. In this paper, we propose a 2D model of overlimiting ion transfer in an ED membrane channel with taking into account both forced and current-induced convection in the absence of chemical effects. The model is based on the fully coupled Navier–Stokes and Nernst–Planck–Poisson equations and does not contain any adjustable parameters. The numerical solution is found without *a priori* separation of the entire solution into electroneutral and charged regions. While the NPP/NS equations are largely used in electrokinetics and nano/microfluidics [30,56,57], certain peculiarities of the proposed model, such as specific boundary conditions, distinguish our model from those developed earlier. Partial solutions of the problem under study and some experimental data are used for assessing the numerical solution.

2. Theoretical

2.1. Governing equations

Consider the system sketched in Fig. 1. Let x and y be the transverse and longitudinal coordinates, respectively; $x=0$ relates to the conditional AEM/solution boundary, and $x=h$ to the boundary with the CEM; $y=0$ corresponds to the inlet and $y=L$, to the outlet of the channel.

In the absence of chemical reactions, the governing equations for a 1:1 electrolyte are written as follows:

$$\vec{j}_i = \pm \frac{F}{RT} D_i c_i \vec{E} - D_i \nabla c_i + c_i \vec{V}, \quad i = 1, 2 \quad (3)$$

$$\frac{\partial c_i}{\partial t} = -\nabla \cdot \vec{j}_i, \quad i = 1, 2 \quad (4)$$

$$\epsilon \epsilon_0 \Delta \varphi = -F(c_1 - c_2) \quad (5)$$

$$\vec{i} = F(\vec{j}_1 - \vec{j}_2) \quad (6)$$

$$\frac{\partial \vec{V}}{\partial t} + (\vec{V} \cdot \nabla) \vec{V} = -\frac{1}{\rho} \nabla P + \nu \Delta \vec{V} - \frac{F}{\rho} (c_1 - c_2) \nabla \varphi, \quad (7)$$

$$\nabla \cdot \vec{V} = 0 \quad (8)$$

where c_i , D_i and $\vec{j}_i$ are the concentration, the diffusion coefficient, and the flux density of ionic species “ i ”, respectively, $i=1$ (cation) or 2 (anion); φ is the electric potential; $\vec{V}$ is the velocity of solution flow; ρ is the solution density (assumed constant); P is the pressure; ϵ_0 is the vacuum permittivity, ϵ is the solution relative permittivity; t is the time; ν is the kinematic viscosity; F , R and T have their usual meanings.

Here P , $\vec{V}$, φ , $\vec{i}$, $\vec{j}_1$, $\vec{j}_2$, c_1 and c_2 are unknown functions of t , x and y . The Nernst–Planck [Eq. (3)] together with the matter conservation law [Eq. (4)], Poisson [Eq. (5)] and electric current

flow [Eq. (6)] equations describe the concentration and potential fields, while the Navier–Stokes [Eq. (7)] and continuity [Eq. (8)] equations describe the velocity field under the action of the forced flow and the electric body force. Two groups of equations are coupled via the velocity present in Eq. (3) and the electric potential and concentrations determining the body force, $\vec{f} = -F(c_1 - c_2)\nabla\varphi$, in Eq. (7).

2.2. Boundary conditions

To find unknown functions listed above, an electrical regime should be specified. We consider the potentiodynamic mode where the potential drop in the system is assumed to be known as a function of time. In addition, boundary conditions on concentrations and fluxes are to be set at the left-hand ($x=0$) and the right-hand ($x=h$) boundaries as well as at the inlet ($y=0$) and the outlet ($y=L$) of the diluting compartment.

2.2.1. Membrane/solution interfaces

Schematically, concentration profiles are shown in Fig. 1. In the bulk of the diluting compartment, the LEN condition is fulfilled. When approaching the interface, the counterion concentration passes through a minimum (denoted as c_{2s} at the AEM surface, and c_{1s} at the CEM surface, respectively), separating the extended SCR and the quasi-equilibrium EDL [17,23]. Within the EDL, the counterion concentration rapidly increases and reaches the concentration of the fixed ions in the membrane bulk. The co-ion concentration exponentially verges towards zero when approaching the (“ideally selective”) membrane. As the membrane surface is inhomogeneous at a nanometer scale, the counterion concentration at the physical boundary (c_{1m} for the CEM) will be a function of the position. However, we can say that c_{1m} should be lower than the fixed ions concentration ($\bar{c}_1$), but of the same order of magnitude. To find c_{1m} , several models could be applied. If the model of homogeneous “charged sponge” [58] is used, we have to assume global electroneutrality at the interface: the positive space charge in the solution near a CEM should be balanced by the negative charge of fixed ions partially bared within the EDL of the membrane, in accordance to the Teorell–Meyer–Sievers model. When using a “space charge model” [59], the membrane surface should be present as a charged wall with a specified surface charge (which can be evaluated when knowing the average fixed ions concentration and the membrane porosity). In this case, the Gouy–Chapman description of the double layer with the NP equation may be applied. However, in the case of 1D modeling [23], the value of c_{1m} does not affect essentially the distribution of concentrations and the potential in the extended SCR. On the other hand, the exact evaluation of c_{1m} does not present the aim of this article. Hence, following Rubinstein and Shtilman [17], we will set the counterion concentration (c_{1m} at the CEM and c_{2m} at the AEM) in boundary conditions (the Dirichlet–Rubinstein boundary conditions), and consider it as a parameter. Note that the same boundary condition was used by Pham et al. [53] and Kwak et al. [30] at the membrane surface.

In the “ideally selective” membrane considered here, the current is carried solely by counterions, hence, at the solution/membrane interface, the counterion partial current density is equal to the total current density and the co-ion normal flux is zero. In addition, we will assume the no-slip condition at the membrane surface. Hence, we have

$$\begin{aligned} \left(\frac{\partial c_1}{\partial x} + \frac{F}{RT} z_1 c_1 \frac{\partial \varphi}{\partial x}\right)(0, y, t) &= 0, \quad \left(\frac{\partial c_2}{\partial x} + \frac{F}{RT} z_2 c_2 \frac{\partial \varphi}{\partial x}\right)(0, y, t) \\ &= -\frac{i_x(0, y, t)}{D_2 F}, \quad c_2(0, y, t) = c_{2m} = N_a c_0, \\ V_x(0, y, t) &= 0, \quad V_y(0, y, t) = 0 \end{aligned} \quad (9)$$

$$\begin{aligned} \left(\frac{\partial c_1}{\partial x} + \frac{F}{RT} z_1 c_1 \frac{\partial \varphi}{\partial x}\right)(h, y, t) &= -\frac{i_x(h, y, t)}{D_1 F}, \quad \left(\frac{\partial c_2}{\partial x} + \frac{F}{RT} z_2 c_2 \frac{\partial \varphi}{\partial x}\right)(h, y, t) = 0, \\ c_1(h, y, t) &= c_{1m} = N_c c_0, \quad V_x(h, y, t) = 0, \quad V_y(h, y, t) = 0 \end{aligned} \quad (10)$$

where N_a and N_c show how many times c_{2m} and c_{1m} are higher than the inlet electrolyte concentration (c_0), respectively.

At $x=0$ and $x=h$, we can also set the values of the electric potential as functions of the local current density. For that, we should specify the PD between the middle of the membranes or the middle of the concentrating compartments. In the first case, we assume that there is no longitudinal potential gradient and no longitudinal electric current in the middle of the membranes. In the second case, the same assumption relates to the middle of the concentrating compartments. Physically, both assumptions hold, and the second one is usually used when applying NP/NS equations with the LEN condition [11,12,37]. Hence, we prefer the second one, also because in this case the PD may be measured experimentally [22,24] by using two Luggin capillaries introduced in the middle of the concentrating compartments (shown by 1 and 1' in Fig. 1). In this case, the total PD in the ED cell pair ($\Delta\varphi_{tot}$) is presented as the sum of three contributions:

$$\Delta\varphi_{tot} = \Delta\varphi_a + \Delta\varphi_{DC} + \Delta\varphi_c \quad (11)$$

where $\Delta\varphi_{DC}$ is the PD within the diluting compartment limited by $x=0$ at the left and $x=h$ at the right, $\Delta\varphi_a$ is the PD across the AEM and a half of the concentrating compartment, $\Delta\varphi_c$ is a similar magnitude for the CEM half-cell (Fig. 3).

To find $\varphi(0, y, t)$ and $\varphi(h, y, t)$ as functions of the normal local current density (i_x) and $\Delta\varphi_{tot}$, we assume that within the membrane EDL, the distribution of counterion is quasi-equilibrium [23], hence, satisfies the Boltzmann law, and the PD is determined by the Donnan equation. The same assumption is accepted for the ‘enriched solution/AEM’ and the ‘CEM/enriched solution’ interfaces. Moreover, for the sake of simplicity, we will not take into account concentration polarization at the membrane interfaces with enriched solution, hence, everywhere in the concentrating compartment the concentration is set equal to c_0 . Under these assumptions,

$$\begin{aligned} \Delta\varphi_a &= -\frac{1}{2} R_{ms} i_x(0, y, t) - \frac{RT}{z_2 F} \ln \frac{c_{2m}}{c_0} = -\frac{1}{2} R_{ms} i_x(0, y, t) \\ &+ \frac{RT}{F} \ln N_a \end{aligned} \quad (12)$$

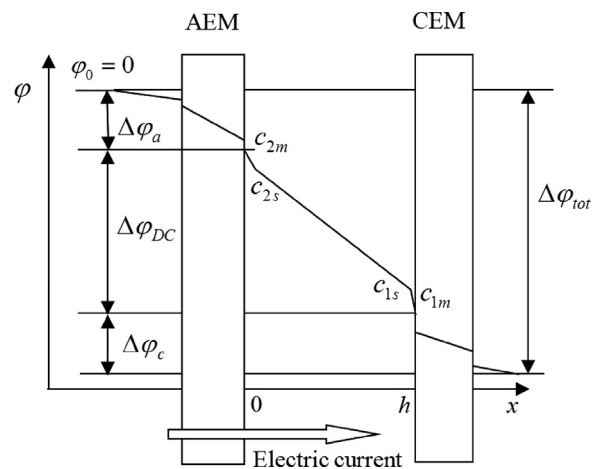


Fig. 3. Schematic distribution of the electric potential. The variations within quasi-equilibrium interfacial EDLs are shown as vertical lines. c_{2s} and c_{1s} relate to the positions of counterion minimum concentration near the AEM and the CEM, respectively; c_{1m} and c_{2m} show the position of the points where the counterion concentration profile passes from the solution to the membrane phase.

$$\Delta\varphi_c = -\frac{1}{2}R_{ms}i_x(h, y, t) - \frac{RT}{z_1F} \ln \frac{c_0}{c_{1m}} = -\frac{1}{2}R_{ms}i_x(h, y, t) + \frac{RT}{F} \ln N_c \quad (13)$$

where R_{ms} is the total ohmic (membrane+solution) resistance of the system involving an AEM and a CEM as well as a concentrating compartment: $-(1/2)R_{ms}i(0, y, t)$ and $-(1/2)R_{ms}i(h, y, t)$ are the ohmic PDs in the corresponding parts of the system under the assumption that the resistance of AEM and that of CEM are the same. The second terms in the left-hand side of Eqs. (12) and (13) involve the total interfacial PD. Thus, for the CEM, we have

$$\Delta\varphi_c^{Don} = -\frac{RT}{z_1F} \ln \frac{\bar{c}_1}{c_{1m}} - \frac{RT}{z_1F} \ln \frac{c_0}{\bar{c}_1} = -\frac{RT}{F} \ln \frac{c_0}{c_{1m}} = \frac{RT}{F} \ln N_c \quad (14)$$

Setting $\varphi=0$ in the middle of the left-hand concentrating compartment, and taking into account that $\varphi(0, y, t) = \Delta\varphi_a$ and $\Delta\varphi_{tot} = \varphi(h, y, t) + \Delta\varphi_c$, we find from Eqs. (12) and (13)

$$\varphi(0, y, t) = \frac{RT}{F} \ln N_a - \frac{1}{2}R_{ms}i_x(0, y, t), \quad \varphi(h, y, t) = \Delta\varphi_{tot} - \frac{RT}{F} \ln N_c + \frac{1}{2}R_{ms}i_x(h, y, t) \quad (15)$$

In computations, we assume $\Delta\varphi_{tot}$ increasing linearly with time:

$$\Delta\varphi_{tot} = -\alpha t \quad (16)$$

where in most runs, $\alpha=0.01$ (V s⁻¹), hence, $\Delta\varphi_{tot}=1$ V is reached after 100 s after switching on the current.

2.2.2. Inlet, outlet and initial conditions

Inlet: $y=0$. We set the inlet feed concentration equal to c_0 at all x except $x=0+$ and $x=h-$, where the counterion concentration is equal to $c_{2m}=N_a c_0$ and $c_{1m}=N_c c_0$, respectively. As at $0 < x < h$ the concentration is uniformly distributed along x , only ohmic resistance occurs and the potential is linearly distributed: $\varphi(x, 0, t) = \varphi(0+, 0, t) + [\varphi(h-, 0, t) - \varphi(0+, 0, t)]x/h$. The PD difference between points $x=0$ and $x=0+$, and $x=h-$ and $x=h$ is described by the Donnan relations. The velocity profile at the inlet is parabolic and satisfies Poiseuille's law, hence not disturbed by electroconvection. Thus, we have:

$$c_i(x, 0, t) = c_0, i = 1, 2; \quad x \neq 0, x \neq h; \quad c_2(0, 0, t) = N_a c_0, c_1(h, 0, t) = N_c c_0; \quad (17)$$

$$\varphi(x, 0, t) = -\frac{1}{2}R_{ms}i_x(x, 0, t) + [\Delta\varphi_{tot} + R_{ms}i_x(x, 0, t)]\frac{x}{h}, \quad 0 < x < h; \quad (18)$$

$$\varphi(0, 0, t) = \frac{RT}{F} \ln N_a - \frac{1}{2}R_{ms}i_x(0, 0, t), \quad \varphi(h, 0, t) = \Delta\varphi_{tot} - \frac{RT}{F} \ln N_c + \frac{1}{2}R_{ms}i_x(h, 0, t) \quad (19)$$

$$V_x(x, 0, t) = 0, \quad V_y(x, 0, t) = 6V_0 \frac{x}{h} \left(1 - \frac{x}{h}\right) \quad (20)$$

Outlet: $y=L$. Here we use “soft” conditions: the velocity distribution is again parabolic and the tangential derivatives of concentrations and potential are set to be zero. These conditions correspond to classical 2D models applying the LEN assumption where no current-induced convection occurs [11,12].

$$\frac{\partial c_i}{\partial y}(x, L, t) = 0, \quad \frac{\partial \varphi}{\partial y}(x, L, t) = 0, \quad (21)$$

$$V_x(x, L, t) = 0, \quad V_y(x, L, t) = 6V_0 \frac{x}{h} \left(1 - \frac{x}{h}\right), \quad (22)$$

Initial conditions: $t=0$. The concentration distribution is set uniform (except at $x=0$ and $x=h$); the initial conditions are

consistent with the other boundary conditions.

$$c_i(x, y, 0) = c_0, i = 1, 2; \quad 0 < x < h; \quad c_2(0, y, 0) = N_a c_0, \quad c_1(h, y, 0) = N_c c_0; \quad (23)$$

$$\varphi(x, y, 0) = -\frac{1}{2}R_{ms}i_x(x, y, 0) + [\Delta\varphi_{tot} + R_{ms}i_x(x, y, 0)]\frac{x}{h}, \quad 0 < x < h \quad (24)$$

$$\varphi(0, y, 0) = \frac{RT}{F} \ln N_a - \frac{1}{2}R_{ms}i_x(0, y, 0), \quad \varphi(h, y, 0) = \Delta\varphi_{tot} - \frac{RT}{F} \ln N_c + \frac{1}{2}R_{ms}i_x(h, y, 0) \quad (25)$$

$$V_x(x, y, 0) = 0, \quad V_y(x, y, 0) = 6V_0 \frac{x}{h} \left(1 - \frac{x}{h}\right) \quad (26)$$

Eqs. (1–26) were solved by applying the finite element method. Within each time step, the Nernst–Planck–Poisson [Eq. (3)] and the Navier–Stokes equations [Eq. (7)] are solved iteratively. When knowing a velocity field from the initial condition [Eq. (26)] or from the previous iteration, the Nernst–Planck–Poisson equations are solved simultaneously. The ion concentrations and electric potential obtained are used to calculate the electrical body force involved in the Navier–Stokes equations [Eq. (7)]. These equations are then solved to obtain the velocity field for the next iteration. The process is repeated until convergence is reached.

The formulation of the mathematical problem in this paper is similar to that applied by Kwak et al. [30]. The difference, in particular, is in the boundary conditions: in Ref. [30] the potential difference $\Delta\varphi_{tot}$ determining the electric regime in the system is set between the anion-exchange and the cation-exchange depleted surfaces. In our model, $\Delta\varphi_{tot}$ is set between the points in the middle of concentrating compartments that allows the direct comparison of calculations with experimental I – V curves measured with help of probing electrodes put in the concentrating compartments neighbor to the desalting channel under study. In addition, we set the PD as a slow function of time that allows us to simulate I – V curves in a similar way as in experiment, and to obtain characteristic oscillations as functions of applied voltage.

3. Results and discussion

In this paper, we are interested in describing the electroconvective mechanism of overlimiting transfer and focusing on the features conditioned by the presence of forced convection. Hence, we will not consider other possible mechanisms: water splitting and accompanying effects, such as Kharkats' exaltation [20] or current induced membrane discharge (CIMD) [33]. The CIMD effect is due to water splitting: in the case of anion-exchange membrane with weakly basic functional groups, OH⁻ ions generated at high currents at the depleted membrane interface may deprotonate the charged (protonated) form of these groups, which results in membrane discharge. As a consequence of this discharge, the membrane permeability towards salt co-ions (entering from the concentrated solution) increases. The appearance of water splitting products as well as salt co-ions in the depleted boundary solution may partially suppress the nonequilibrium extended space charge and electroosmotic flow of the second kind studied here.

The effects mentioned above are important and pertinent to the overlimiting current phenomenon. However, only the effects, which directly relate to the interaction between electric current and volume flow via electric space charge, are considered in this study.

3.1. Current–voltage curve

The results of simulation presented here are obtained for a flow-through ED cell (Fig. 1) in the case of dilute NaCl solutions. The inlet concentration, 10^{-4} and 10^{-5} M, was taken in order to benefit of a higher EDL thickness near the membranes for the numerical calculations: since a part of the EDL is taken into account within the interval $0 \leq x \leq h$, it is important to have here no too rapidly changing functions. The nonuniform grid used in the numerical calculations contained at least ten points inside the EDL. As the diffusion coefficient of the chloride ion is about 1.5 times higher than that of sodium, the limiting current density, according to Eq. (1), is about 1.5 times higher at the AEM in comparison with the CEM. Hence, with increasing PD, the phenomena accompanying concentration polarization occur firstly at the CEM, and only under essentially higher PD, at the AEM.

As we can see in Fig. 2, the “basic” model gives qualitatively correct description of I – V curve for ED membrane cells with forced convection. Both the experimental and the theoretical curves have a linear initial part. Then, after an intermediary non-linear under-limiting region, the current grows over the limiting value, i_{lim} . The latter is proper to “classical” 2D models [11,12]: the current density value is really limited by i_{lim} . There is a sloping plateau where the current density smoothly increases with increasing voltage. Then the plateau changes for a steeper region where oscillations of current density occur. Note that (at least for short-length channels in which the concentration decrease in the channel bulk is negligible) the point of intersection of the tangents drawn to the initial part and to the sloping plateau of the curve is very close to i_{lim} , which can be calculated using L  v  que's Eq. (2).

No electroconvection occurs at under-limiting current densities. It starts when concentration polarization becomes sufficiently high. Calculations made for points located in the range of 5–100 μm (while setting $R_{ms}=0$) from the membrane surface in the section situated at 1 mm from the channel entrance show (Fig. 4) that when the PD attains a value close to 0.2 V (the ohmic component is omitted) the diffusion ($i_{1dif} = -FD_1\partial c_1/\partial x$) and the migration ($i_{1migr} = -(F^2/RT)D_1\partial\varphi/\partial x$) components of the overall counterion current density normal to the surface attain their “limiting” values while the convective component ($i_{1conv} = Fc_1V_x$) remains zero. Note that this PD corresponds approximately to the situation where the electrolyte concentration at the membrane (c_s^l) reaches a value about $0.01c_0$ (becoming negligible in comparison to c_0). In this case, the PD across the depleted DBL is about $2(RT/F)\ln(c_0/c_s^l) \approx 0.2$ V [38]; the other components of the overall PD are smaller. With a further increase of PD above 0.2 V up to approximately 1 V, the values of i_{1dif} and i_{1migr} do not change while i_{1conv} monotonically grows due to stable electroconvective vortices. This regime relates to the smooth sloping plateau region of the I – V

curve, which goes on until PD reaches a threshold close to 1 V. Then all three components start to regularly oscillate with a period of the order of 1 s, and after a second threshold the oscillations become not regular.

3.2. Electroconvection modes

Though there is a similarity in the phases of development of electroconvection in cells with and without forced convection, there are also essential differences. As we noted in the introduction section, the distribution of electric field should be inhomogeneous to produce electroconvective vortices. But how this inhomogeneity can be attained in a cell with smooth homogeneous membranes? In the case of no forced convection and homogeneous membranes, the current lines are uniform and normal to the membrane surface as well as the electric body force. The latter is counterbalanced by a pressure gradient. However, the system is not stable, as it was shown by Rubinstein and Zaltzman [18]. Beyond a certain threshold, any fluctuation in concentration or velocity is sufficient to initiate the process of vortices generation. The presence of vortices assures inhomogeneous distribution of electric field necessary for reproducing electroconvective motion.

In the systems with curved or electrically heterogeneous surface, tangential component of electric current flowing within the space charge region can produce electroconvection, as it was shown by Dukhin and Mishchuk [41,43–45]. In this case, a stable electro-osmotic flow of the second kind occurs and no instability is needed for the onset of electroconvection. These two modes of electroconvection are recognized in literature [31,48] and sometimes are shortly called as Dukhin's and Rubinstein's electrokinetic modes [60].

In the case of a cell with forced flow, concentration distribution is not uniform in the longitudinal direction: the electrolyte concentration decreases while the diffusion layer thickness increases with the distance from the inlet. As a result, the solution resistance increases with y , and the current lines are denser near the inlet. There is a tangential component of the current/electric field (Fig. 5) and a tangential body force $f_y = -F(c_1 - c_2)\partial\varphi/\partial y$ both directed towards the inlet (Fig. 6). This force produces steady-state electroconvective clockwise vortices. The tangential force is not zero even at 0.2 V. However, it is located at quite small distance from the membrane, $< 1 \mu\text{m}$ (Fig. 6a), where the viscous forces are important. Thus, its contribution into the enhancement of current becomes noticeable at 0.25–0.3 V per cell pair (without taking into account the ohmic resistance of the membranes and solution) when the current calculated with NPP/NS equations becomes higher than that calculated by using NP/NS (Figs. 2 and 4).

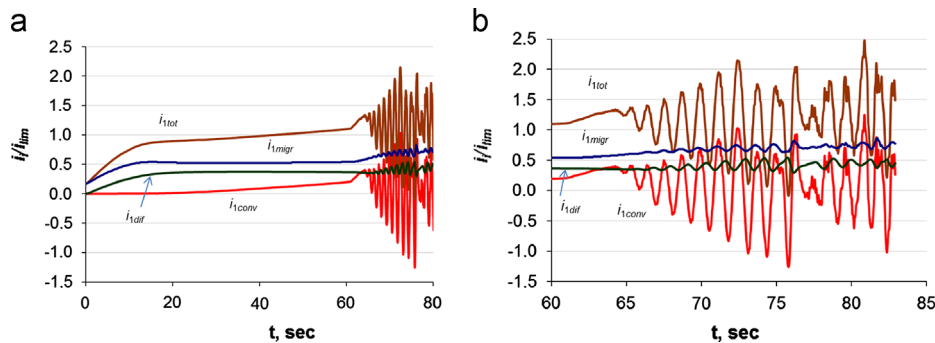


Fig. 4. Diffusion ($i_{1dif} = -FD_1\partial c_1/\partial x$), migration ($i_{1migr} = -(F^2/RT)D_1\partial\varphi/\partial x$) and convective ($i_{1conv} = Fc_1V_x$) components of the overall current density (i_{1tot}) divided to the limiting current density i_{lim} vs. the time t . PD applied to a cell pair (in V) increases with t (in sec) as $\Delta\varphi_{tot} = 0.02 t$. The calculations are made for a point situated at 0.1 mm from the right boundary (CEM) and 1 mm from the entrance of the desalting channel, $c_0 = 10^{-4}$ M NaCl, $h = 1$ mm, $L = 2$ mm, $V_0 = 0.8 \text{ mm s}^{-1}$. **b** is a magnification of **a**.

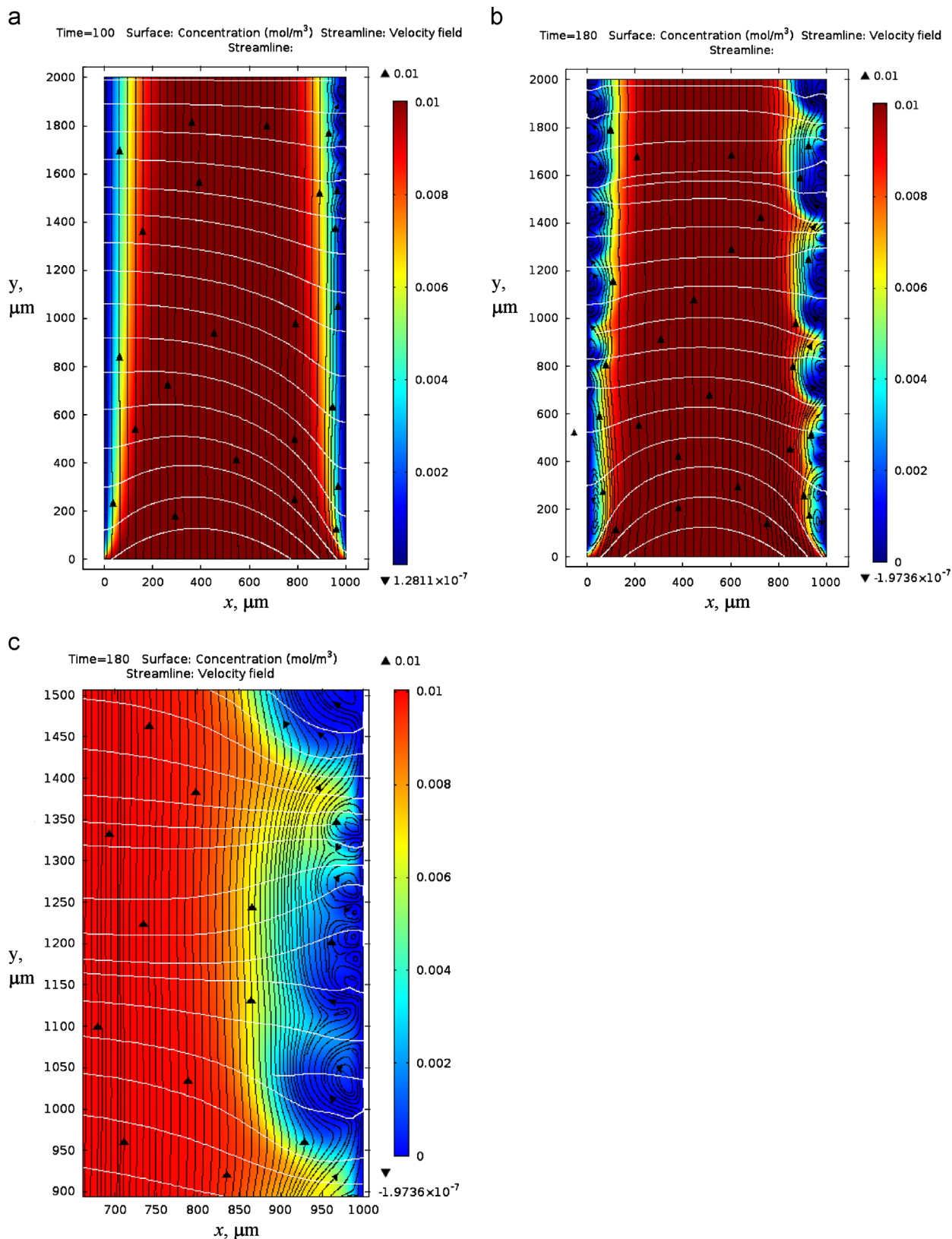


Fig. 5. Distribution of electric current (white lines) and fluid velocity streamlines at two different voltages (times), $\Delta\phi=1.0$ V (a) and $\Delta\phi=1.8$ V (b). The first picture corresponds to emergence of unsteady vortices at the CEM near the outlet, while a long vortex near the inlet remains stable. The second picture shows unsteady vortices at both membranes. c is the magnification of b. The parameters used in calculation are the same as in Fig. 2.

Note that f_y is non-zero approximately in the same region as the extended space charge $\tilde{\rho}$ (Fig. 6a and b). However, the local minimum of f_y is at a higher distance from the membrane

in comparison with the local maximum of $\tilde{\rho}$. The tendency of f_y to decrease when approaching $x=h$ is apparently due to the boundary condition setting a constant value of the electric

potential ($\Delta\varphi_{\text{tot}}$) in the middle of the right-hand concentrating compartment.

In the range of 0.3–0.95 V, there is no oscillations on the I - V curve while the current density is higher than the limiting one. The electrolyte concentration at the cation-exchange membrane is more than 100 times lower than in the bulk (see the estimations in Section 3.1), and an extended space charge region occurs (Fig. 6b). Thus, in these conditions, we have electroconvection occurring as steady-state electro-osmotic flow of the second kind. In particular, in the section $y=400\text{ }\mu\text{m}$, the tangential body force is stable and negative in this range of voltage (Fig. 6a). The electrokinetic mode observed in this case is similar to that known as the Dukhin–Mishchuk one. However, there is an important difference: no membrane surface undulation/nonflatness or electric heterogeneity takes place in our case. The tangential electric field needed to realize this electroconvective mode occurs due to a non-uniform distribution of the electrolyte concentration and the DBL thickness along the membrane channel.

The stationary vortices occur not far from the channel inlet. They are difficult to see in the full-scale animation of velocity stream-lines (Video 1), as the deviation of the velocity from the initial Poiseuille distribution is low and occurs at a very small distance from the membrane. Note, however, that the contribution of the $c_1 \vec{V}$ term in the Nernst–Planck equation is quite important – because of a small value of the diffusion coefficient. Namely, this term is responsible for more than one-and-a-half-fold increase of the current in the inclined plateau region of the I - V curve, Fig. 2. The perturbations of velocity are well seen in the velocity field graph in the Video 1 c and d. A slight deviation from zero of the normal velocity near the cation-exchange membrane is already seen at 0.4 V (for the conditions specified in the video legend) in the section at $400\text{ }\mu\text{m}$ from the inlet, while similar deviations in the $1900\text{ }\mu\text{m}$ section are seen only at about 0.6 V. The normal and tangential velocities at $400\text{ }\mu\text{m}$ remain steady up to 1.35 V, while at $1900\text{ }\mu\text{m}$ the threshold of passage to unsteady state is at about 0.95 V. The tangential velocity just near the membrane surface at $400\text{ }\mu\text{m}$ from the inlet in the range 0.4–1.35 V is negative, that coincides in sign with the tangential component of the current (Fig. 5a) and the electric body force (Fig. 6a). A long stable vortex turns clockwise near the CEM close to the inlet in this range of voltage. At higher voltages ($> 1.35\text{ V}$), instability in vortex structure near the channel inlet occurs, and the tangential body force changes its sign with time. For example, at 1.8 V at $y=400\text{ }\mu\text{m}$ and the distance about 6.3 mm from the CEM, $f_y \approx +2.8 \times 10^4\text{ N m}^{-3}$ that is one order of magnitude higher than the values in steady state shown in Fig. 6a.

Supplementary material related to this article can be found online at <http://dx.doi.org/10.1016/j.memsci.2013.07.033>.

Above 0.95 V (the conditions specified in the legend of the Video 1), oscillations appear on the I - V curve. We have the behavior known as the Rubinstein–Zaltzman electroconvective instability. First one pair, then more, of small vortices emerge at the CEM near the exit of the channel. At the beginning, the vortices appear and disappear regularly producing regular oscillations of current density on the I - V curve (Fig. 2b). With increasing voltage, the situation becomes more and more complicated: the vortices interact between them and with the forced flow; within big vortices, the smaller ones emerge (Video 1). The electric current oscillations become more and more irregular. The essential difference observed in the channels with forced convection in comparison to a no-forced-flow system studied by Rubinstein and Zaltzman [18] is that the symmetry of vortex pairs is broken by the forced flow. On the CEM, the clockwise rotating vortices are significantly larger than the counterclockwise ones (Fig. 5c); on the AEM, the counterclockwise rotating vortices are larger than the clockwise ones. In both cases, the forced (shear) flow is

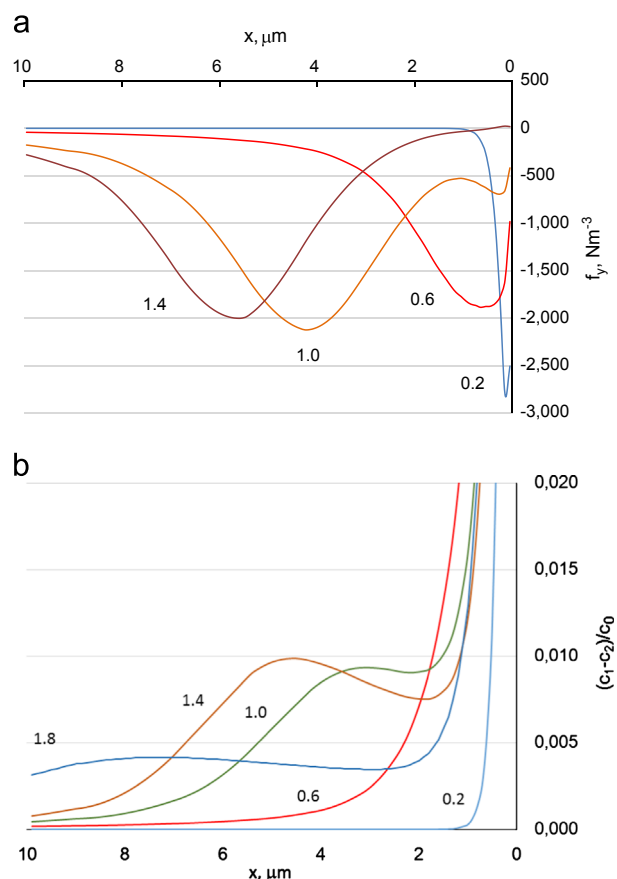


Fig. 6. Profiles of the tangential electric body force, f_y , (a) and the dimensionless space charge density, $\tilde{\rho} = (c_1 - c_2)/c_0$, (b) vs. the distance from the CEM in the cross-section at distance $y=400\text{ }\mu\text{m}$ from the channel inlet, at different PD (shown in V near the curves) across the cell pair. Other parameters are specified in Fig. 2.

favorable to the vortices, the rotational velocity of which in the channel core matches the velocity direction of the forced flow; the vortices rotating in the opposite direction are partially suppressed. This result agrees well with that by Kwak et al. [30].

The singularity of the behavior of the system with forced flow is also in the fact that steady-state and unsteady vortices are observed simultaneously. As we have mentioned above, the voltage threshold of unsteady vortices appearance is about 0.95 V. First they appear near the channel outlet (Video 1). Then inside the large long vortex near the channel inlet, several small ones appears. They are emerging from the long one and traveling towards the outlet while increasing in size (Fig. 5b). Small vortices are also seen in other relatively large vortices. While these oscillations in velocity near the CEM appear in the $1900\text{ }\mu\text{m}$ section at about 0.95 V, in the $400\text{ }\mu\text{m}$ section they appear only at 1.35 V. The situation near the AEM is similar: first, at about 1.5 V, the velocity oscillations appear near the channel exit (Video 1, the $1900\text{ }\mu\text{m}$ section), then at nearly 1.8 V, near the channel inlet (the $400\text{ }\mu\text{m}$ section in Video 1). Each event (appearing of new type of oscillations) has a response in the I - V curve via a new steeper slope.

The described scenario agrees with that visualized by Kwak et al. [29] in a microscale ED channel. In the video made by these authors, a relatively long vortex is seen near the entrance of the DC. It produces smaller vortices with a certain regularity at relatively low voltages; these vortices travel towards the exit while increasing in size. The generation of vortices near the AEM occurs at higher voltages. When the intensity of vortices generation becomes sufficiently high as well as their size, the situation becomes irregular.

In a more recent publication, Kwak et al. [30] have found that the ratio of the EC zone thickness (d_{ec}) ($\approx$ vortex diameter) to the distance between the membranes (h) is proportional to $(\Delta\varphi_{ec})^{2/3}V_0^{-1/3}$, where $\Delta\varphi_{ec}$ is the potential drop in the EC zone, and V_0 is the average forced flow velocity. Our simulations approximately confirm this finding. Really, for a channel with $h=1$ mm and $L=8$ mm, under a fixed potential drop (0.5 V over the cell) and $(V_0)_1=0.1$ mm s⁻¹, the dimensionless EC zone thickness, $(d_{ec}/h)_1$, is found about 0.2, while at $(V_0)_2=0.01$ mm s⁻¹ and the same voltage, this thickness, $(d_{ec}/h)_2$, is about 0.45. The ratio of $(d_{ec}/h)_2$ to $(d_{ec}/h)_1$ is really close to $[(V_0)_1/(V_0)_2]^{1/3}=10^{1/3}$.

To study the dependence of d_{ec} on the potential drop, we have presented d_{ec}/h as a function of $(\Delta\varphi_{ec})^{2/3}$ under a fixed $V_0=0.8$ mm s⁻¹ and other channel parameters specified in Fig. 2. With an accuracy of approximation $R^2=0.98$, we have found that $d_{ec}/h=0.0196[(\Delta\varphi_{ec})^{2/3}V_0^{-1/3}]-0.1084$, which is rather close to the experimental data by Kwak et al. [30]: $d_{ec}/h=0.010(\Delta\varphi_{ec})^{2/3}V_0^{-1/3}-0.171$. The overall potential drop in our simulations varied up to 1.8, while the calculated potential drop over the EC zone changed from 0.41 to 0.78 V, and the EC zone thickness, from 6 to 70 μ m.

The Video 2 made at the Voronezh State University with help of laser interferometry technique developed by Shaposhnik et al. [26,27] shows concentration profiles oscillating with time obtained in an ED channel at an current density approximately equal to $8i_{lim}$, the PD applied per a cell pair is 8 V. Video 3 shows a similar video at $9i_{lim}$ and 10 V. The oscillations of concentration profile, which are apparently due to appearance and disappearance of electroconvective vortices in the vicinity of the point of observation, are well seen.

Supplementary material related to this article can be found online at <http://dx.doi.org/10.1016/j.memsci.2013.07.033>.

3.3. Oscillations

Fig. 7 shows experimental chronopotentiograms of a Nafion®-117 membrane obtained under a fixed current at sufficiently large time when concentration profiles are established. These curves show that there are several typical periods of oscillation, which are seen also in the calculated curves shown in Fig. 4. The dominant ones are quasi-regular oscillations of the period close to 1 s. These oscillations seem due to appearance/disappearance of an electroconvective vortex in the vicinity of the point where probing is realized. It can be seen from Videos 2 and 3 that the main period of profile oscillation is close to 1 s as well. Appearance/disappearance of electroconvective vortices in an ED channel with a period of the order of 1 s was also visualized by Kwak et al [29]. In addition to this main period, there is a larger period of oscillations counting 5–10 s, which is also well seen in Fig. 4. These oscillations should be due to convective transfer of vortices with the forced flow. There are also oscillations with lower periods, of the order of 0.1 s and less. They might be caused by smaller vortices emerging within big ones (Video 1, $t > 170$ s).

The amplitude of measured PD oscillations strongly depends on the distance from the probing point to the membrane surface. In the experiments shown in Fig. 2a and b, a Luggin capillary was used with a beveled tip having an oval cross-section with a transverse diameter of about 1 mm [22]. The center of the capillary tip was at about 1 mm from the membrane surface in the diluting compartment; the second capillary was introduced in the concentrating compartment. The spacing between the membranes in both compartments was 6 mm. The amplitude of PD oscillations depends on the membrane and varies up to 50 mV (for Nafion, Fig. 2b). For other membranes it was lower, between 5 and 15 mV. In calculations, in the unsteady overlimiting region of the I - V curve, we find the amplitude of oscillations (via the amplitude

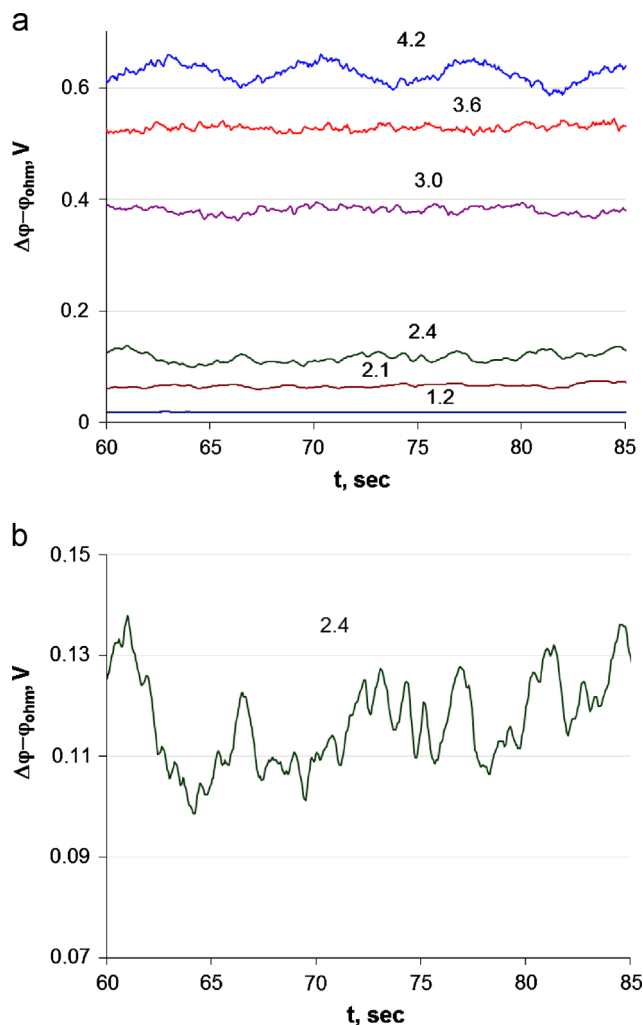


Fig. 7. a and b. Chronopotentiograms of a Nafion®-117 membrane in quasi-steady state obtained at different current densities i . The values of the ratio i/i_{lim} are shown near the curve.

of the current density, which is calculated directly, Fig. 2b) close to 10 mV. However, note that the amplitude of the PD oscillations over a cell pair (where the measuring capillaries are introduced in the concentrating compartments next to the diluting one under study) is essentially lower than in the case where one of the capillary is within the desalting compartment.

3.4. Concentration and electrochemical potential profiles

Fig. 8 shows experimental (obtained using laser interferometry technique [26,27]) and calculated concentration profiles at different ratios i/i_{lim} . The “basic” model gives correct profile shapes. At low currents, the profile is near-linear near the membrane surface, its slope monotonically decreases with increasing distance from the membrane due to increasing contribution of convection to the ion flux. However, at $i > i_{lim}$ the shape of concentration profile is more complicated. An electroconvective vortex formed near the membrane surface contributes to mixing solution that results in decreasing concentration gradient in the zone of vortex. As the vortex occurs near the surface, the zone where diffusion remains important, generally shifts from the surface towards the bulk. Thus, the diffusion layer is no longer the boundary one. The mixing produced by vortices leads to expansion of depletion zone where concentration is lower than in the bulk. Hence, the diffusion layer understood as the zone where the concentration differs from that

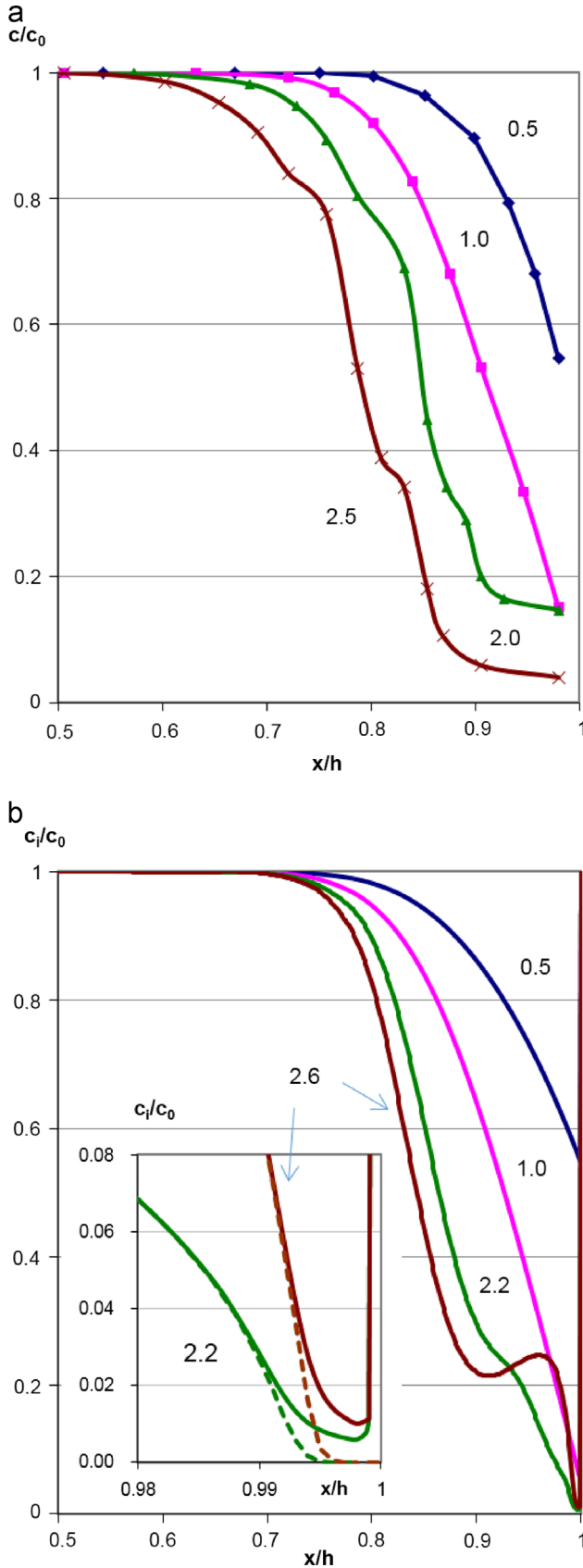


Fig. 8. Concentration profiles found experimentally using laser interferometry for a Russian CEM MK-40 coated with a 15 μm film of Nafion[®] in 0.02 M NaCl (a), and calculated according to the “basic” model (b) for different ratios i/i_{lim} shown near the curves. Insert in b presents calculated profiles of cation (solid curves) and anion (dashed curves) near the boundary. In a, a part of the profile is not seen due to uneven membrane surface.

in the bulk increases with increasing voltage [26,27]. At sufficiently high PD, the profile becomes unsteady: the vortices go up and down, the concentration gradient near the surface varying in opposite direction. Flat regions (as it was also shown by Kwak et al. [29]) and even inversion of concentration profile slope may appear to be then replaced by a steeper slope (Video 1). The period of oscillations is of the order of 1 s. Note that in experiment, the situation is even more complicated: several zones of flat profile can occur (Fig. 8a), apparently due to several vortices simultaneously present in the depletion zone (see Fig. 5c).

The extended SCR seen in the insert in Fig. 8b, has a thickness 5–10 μm at $i > 2i_{\text{lim}}$. The thickness of the equilibrium EDL corresponding to the inlet electrolyte concentration used in this simulation (10^{-5} M) is about 100 nm (the Debye length $L_D = \sqrt{\epsilon\epsilon_0 RT / (2cF^2)}$). It can be seen that the minimum concentration of the counterion at the CEM is about $0.6 \cdot 10^{-7}$ M at $i = 2.2 i_{\text{lim}}$ ($\Delta\varphi_{\text{tot}} = 1.5$ V). It is interesting that at $i = 2.6 i_{\text{lim}}$ this minimum concentration is higher, apparently due to a vortex, which brings fresh solution from the bulk. It is possible to evaluate the thickness of the quasi-equilibrium EDL, where the counterion concentration increases very rapidly tending to the concentration $\bar{c}_1$ in the membrane bulk (Fig. 1): it is about 2 μm if measured as the distance between the point of minimum counterion concentration and the membrane surface. This value is in a good agreement with the value (2.3 μm) calculated by using the equation

$$\lambda = 2L_D \left(\sqrt{\frac{c_0}{c_{1s}}} - \sqrt{\frac{c_0}{c_{1m}}} \right) \quad (27)$$

for $c_{1s} = 0.6 \cdot 10^{-7}$ M and $c_{1m} = 10^{-5}$ M. Eq. (27) is obtained [38] by integration of the Nernst–Planck Eq. (3) written for the counterion (1) within the quasi-equilibrium EDL while neglecting the terms “ j_1 ” and “ $c_1 v$ ” and assuming $c_2 \ll c_1$.

In order to verify how strong is the deviation from equilibrium within the SRC, we have calculated the profiles of dimensionless electrochemical potential (normalized by RT) of the cation near the CEM (Fig. 9),

$$\tilde{\mu}_1 = \ln c_1 + \frac{F}{RT} \varphi \quad (28)$$

at different PD across the cell pair. It can be seen that there is a part of these profiles next to the membrane surface where $\tilde{\mu}_1$ varies quite slightly. This part relates to the quasi-equilibrium EDL where the counterion concentration and the dimensionless space charge density $(c_1 - c_2)/c_0$ sharply increase (Fig. 6b).

Some characteristics of the quasi-equilibrium EDL (defined as the layer between the membrane surface and the point where the counterion concentration attains its minimum, Fig. 8b) and the extended SCR are shown in Table 1. The thickness of the EDL (λ) is

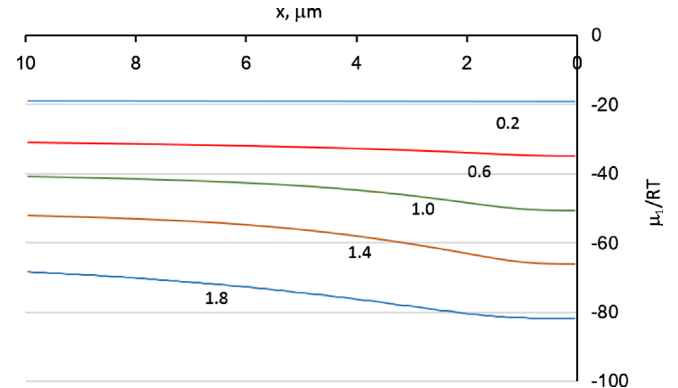


Fig. 9. Dimensionless electrochemical potential of the cation vs. the distance from the CEM in the section $y = 400 \mu\text{m}$ from the channel inlet, at different PD (shown in V near the curves) across the cell pair. Other parameters are specified in Fig. 2.

Table 1

Some characteristics of the equilibrium EDL and the extended SCR on the CEM at different PD in the section $y=400\ \mu\text{m}$ of an ED cell specified in Fig. 2. Calculations by using the basic model and Eq. (27).

$\Delta\varphi_{\text{tot}}$ (V)	i/i_{lim}	Equilibrium EDL thickness (μm)	Equilibrium EDL thickness, μm , Eq. (27)	Extended SCR thickness (μm)	Variation of $\tilde{\mu}_1$ in equilibrium EDL	Variation of $\tilde{\mu}_1$ in extended SCR
0.2	0.8	0.7	0.1	No	0.019	
0.6	1.4	1.6	1.0	7	0.72	3.2
1.0	1.8	2.0	1.8	10	2.3	9.8
1.4	2.1	1.9	2.0	> 10	2.9	> 14
1.8	2.8	3.0	3.1	> 10	3.6	> 13

Table 2

Potential drop in the equilibrium EDL on the CEM in the section $y=400\ \mu\text{m}$ of an ED cell specified in Fig. 2.

$\Delta\varphi_{\text{tot}}$ (V)	PD calculated numerically (mV)	PD calculated by using Eq. (29) (mV)
0.2	19.2	19.0
0.6	108	91
1.0	179	120
1.4	198	125
1.8	235	145

found numerically and compared with the value obtained by applying Eq. (27). As it can be seen, at low PD, at $i < i_{\text{lim}}$ ($\Delta\varphi_{\text{tot}}=0.2\ \text{V}$), and at a slight excess over i_{lim} ($\Delta\varphi_{\text{tot}}=0.6\ \text{V}$), the deviation from the equilibrium within the “quasi-equilibrium EDL” evaluated by the variation of $\tilde{\mu}_1$ is weak: it is lower than 1. However, at higher voltages, the deviation becomes noticeable. Note that Eq. (27) is rather well verified at high voltages.

The invariance of the electrochemical potential within the EDL allows application of the Poisson–Boltzmann equation that results in obtaining the Donnan equation:

$$\Delta\varphi_{\text{EDL}} = -\frac{RT}{F} \ln \frac{c_{1m}}{c_{1s}} \quad (29)$$

Table 2 compares the values of PD across the quasi-equilibrium EDL ($\Delta\varphi_{\text{EDL}}$) calculated numerically and by applying Eq. (29). It can be seen that the difference in two values is small for an under-limiting current density ($\Delta\varphi_{\text{tot}}=0.2\ \text{V}$). There is only a slight excess of the value found numerically over that calculated by using the Donnan Eq. (29). However, this excess grows with increasing PD/current density. Evidently, the cause is in the fact that the ohmic contribution to the PD, which is neglected in the derivation of Eq. (29) starting from the Nernst–Planck equation, increases with increasing current.

3.5. Use of the slip condition

The “basic” NPP/NS model gives a qualitatively adequate description of overlimiting transfer stimulated by electroconvection. There is a good agreement with experimental I – V curves and concentration profiles. As there is no fitting parameters, the model gives only one I – V curve for flat membranes while the experiment shows a great variety of shapes of such curves depending on surface inhomogeneity, hydrophobicity, water splitting rate, cell geometry and other [9,21–29,34,36,54]. Further development of the model is needed for better understanding the overlimiting behavior of ED and micro-nanofluidic systems. One of the possible direction of such development might be in taking into account fluid slip occurring at a hydrophobic surface. In this paper, we propose a rather simple way to introduce fluid slip condition via

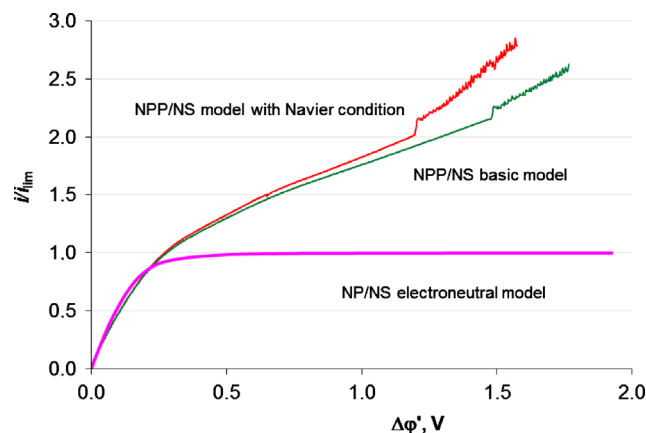


Fig. 10. Comparison of two I – V curves calculated by using the NPP/NS model with the no-slip [Eqs. (9,10)] and the slip [Eq. (30)] conditions, $b=1\ \mu\text{m}$. The other parameters are the same as in Fig. 2.

the Navier equation [61,62]

$$u_{\text{slip}} = b(\partial V_y / \partial n)_{x=h} \quad (30)$$

showing that the slip velocity directed in parallel to the membrane surface, u_{slip} , is the proportional to the normal derivative of the longitudinal velocity at the surface (here, of the CEM); b is the proportional constant known as the slip-length.

When Eq. (30) replaces Eqs. (9) and (10) for V_y in simulation of I – V curve, we obtain a shift of the curve towards higher current densities in the range of sufficiently high voltages (Fig. 10). Thus, this result shows that the fluid slip has an important impact on the membrane electrochemical behavior by reducing viscous restrictions and facilitating formation of electroconvective vortices.

4. Conclusion

In this paper, we propose a “basic” mathematical model of ion transfer in an electrodialysis cell with a cation- and an anion-exchange membranes, which is aimed at the description of intensive current regimes. The model is built on the first principle law equations and involves no empirical assumptions or fitting parameters. A set of NPP/NS fully coupled equations is solved numerically with Dirichlet–Rubinstein boundary conditions in potentiodynamic mode, where the potential difference per ED cell pair is known (generally, as a function of time). The model gives a qualitatively adequate description of overlimiting transfer stimulated by electroconvection. For the first time, a realistic I – V curve having the same main features as the experimental ones is calculated for an ED flow-through cell. A linear initial part of the curve is followed by a sloping plateau situated over the value defined as “limiting” current in the classical theory based on the electroneutrality assumption. The initial part of the plateau is smooth and relates to stable electroconvective vortices occurring due to inhomogeneous distribution of electric current density caused by partial desalination of the solution flowing through the cell. A tangential component of current density passing through an extended space charge region at the membrane surface produces electroconvection of the second kind electroosmosis. This electroconvection mode is similar to that known as the Dukhin–Mishchuk one. However, the difference is in the fact that no membrane surface undulation or electric heterogeneity take place in our case, while it is important for realizing the Dukhin–Mishchuk mode. Onset of unstable electroconvection at higher voltages (the Rubinstein–Zaltzman instability) results in appearance of oscillations on the I – V curve and in increasing its slope.

Initially, unsteady vortices emerge near the outlet of the desalination channel; then, at higher voltages, they occupy the entire channel. In the case of NaCl solution, electroconvective vortices emerge firstly at the cation-exchange membrane, and with increasing voltage, they appear at the anion-exchange membrane where their evolution follows the same scenario as at the cation-exchange membrane. Another singularity of the system is also in the fact that within a certain voltage range, both electroconvective modes occur simultaneously: unsteady vortices emerge near the channel outlet, while near the inlet the vortices remain stable. It is explained by a nonuniform distribution of the DBL thickness δ along the channel: δ rapidly increases at the channel inlet and becomes nearly invariable close to the outlet.

The basic model gives a new insight on the generalities and some important details of electroconvection enhancement of ion transfer in flow-through ED cells. We would like to emphasize that the model contains no adjustable parameters and relates to an ideal ion-exchange membrane. However, there is a great diversity in membrane properties: the membranes differ by the degree of electric/geometric heterogeneity, by the degree of hydrophobicity, by ability to catalyze or not water splitting reaction, by the nature of fixed charged sites and other. There are a number of levers for affecting electroconvection and enhancing ion transfer: the use of profiled surface [9], ion-conductive spacers [63], membranes with suppressed water splitting [25] and other including novel types of water treatment devices [2–5]. We think, the model can be applied to describe these effects, if appropriate new conditions with appropriate parameters are added. The case of hydrodynamic slip is considered as an example of a such possible development. Further theoretical and experimental studies are needed to handle phenomena occurring in intensive current regimes.

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